



Software Quality Metrics

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Agenda

Quality Metrics Definition and Objectives

- Requirements of Quality Metrics

- Types of Quality Metrics

- Definition of Process Metrics

Definition of Product Metrics

What is Quality Metrics?

Definition

- Quantitative measure of the degree to which an item possesses a given quality attribute
- Function whose input is software data and output is a single numerical value —degree to which the software possesses a given quality attribute

• Objectives

- To facilitate management control (for planning and execution of management decisions)
- To identify situations that may need preventive or corrective actions

Quality Metrics - Requirements

General Requirements

- Relevant —related to attribute of importance
- Valid —measures the attribute
- Reliable —produces similar results
- Comprehensive —applicable to variety of situations
- Mutually Exclusive —measures something new

• Operative Requirements

Easy and Simple —Data collection is simple

Do not need independent data collection —Collected along with other project data

- Immunity —Others cannot change or manipulate the data

Types of Software Quality Metrics

Process Metrics

- Related to Software Development Process

- Product Metrics

- Related to Software maintenance

Process Metrics

Types

- Software Process finality Metrics
- Software Process Timetable Metrics
- Error Removal Effectiveness Metrics
- Software Process Productivity Metrics

Process Quality Metrics

- Two Classes
 - Error Density Metrics
 - Error Severity Metrics
- Error Density Metrics
 - Software Volume
 - Use the Lines of Code (LOC) or Function Points
 - Errors Counted Measures
 - Number of Code Errors (NCE)
 - Weighted Number of Code Errors (WCE)

NCE and WCE

Error Severity Class	Relative Weight
Low Severity	1
Medium Severity	3
High Severity	9

Severity Class	# of errors	Relative Weight	Weighted errors
Low	42	1	42
Medium	17	3	51
High	11	9	99
Total	70		192
NCE	70		
WCE			192

Error Density Metrics

Code	Name	Formula
CED	Code Error Density	$CED = \frac{NCE}{KLOC}$
DED	Development Error Density	$DED = \frac{NDE}{KLOC}$
WCED	Weighted Code Error Density	$WCED = \frac{WCE}{KLOC}$
WDED	Weighted Development Error Density	$WDED = \frac{WDE}{KLOC}$
WCEF	Weighted Code Errors per Function Point	$WCEF = \frac{WCE}{NFP}$
WDEF	Weighted Dev. Errors per Function Point	$WDEF = \frac{WDE}{NFP}$

NCE: Number of Code Errors

NDE: Number of Development Errors

WCE: Weighted Number of Code Errors

WDE: Weighted Number of Dev. Errors

KLOC: Thousands Lines of Code

NFP: Number of Function Points

Error Severity Metrics

C'ode		
ASCE	Average Severity of Code Errors	$ASCE = \frac{WCE}{NCE}$
ASDE	Average Severity of Dev. Errors	$ASDE = \frac{WDE}{NDE}$

For cases to identify if the number of severe errors are increasing
With respect to the all other errors in the system

Software Process Timetable Metrics

Based on

- Success (completion of milestones per schedule)
- Failure (non-completion per schedule)

Formula		
TTO	Time Table Observance	$TTO = \frac{MSOT}{MS}$
ADMC	Avg. Delay in Milestone Completion	$A \quad E \quad \frac{TCDAM}{MS}$

MSOT: Milestones Completed on Time

MS: Total Number of Milestones

TCDAM: Total Completion Delays

Error Removal Effectiveness Metrics

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DERE	Dev. Error Removal Effectiveness	$DERE = \frac{NDE}{NDE + NYF}$
DWERE	Dev. Weighted Errors Removal Effectiveness	$DWERE = \frac{WDE}{WDE + WYF}$

NYF: Number of Software Failures detected in a year of Maintenance Service

WYF: Weighted Number of Software Failures detected in a year of Maintenance Service

Software Productivity Metrics

Direct Metrics

Project's Hit Velocity

Indirect Metrics

Software Reuse

Indirectly affects productivity and effectiveness

Code	Name	Formula
DevP	Dev. Productivity	$DevP = \frac{DevH}{KLOC}$
FDevP	Function Point Dev. Productivity	$FDevP = \frac{DenH}{NFP}$
CRe	Code Reuse	$CRe = \frac{ReKLOC}{KLOC}$
DocRe	Doc. Reuse	$DocRe = \frac{ReDoc}{NDoc}$

DevH: Total Hours Spent in Development of Code
 ReKLOC: Number of thousand lines of code reused
 ReDoc: Number of reused pages in Documentation
 NDoc: Number of pages in Documentation

Product Metrics

- Refers to the operational phase of the product
- Related to Customer Service
 - Help Desk Services (HD)
 - Corrective Maintenance Services

Corrective Maintenance Services

- Software System Failure Density Metrics
- Software System Failure Severity Metrics
- Failure or Maintenance Services Metrics
 - Software System Availability Metrics

Failure Density Metrics

- No. or weighted number of failures
- Identified from software maintenance reports

Code	Name	Formula
SSFD	Software System Failure Density	$SSFD = \frac{NYF}{KLMC}$
WSSFD	Weighted SSFD	$FDevP = \frac{WYF}{KLMC}$
WSSFF	Weighted SS Failure per Function Point	$CRe = \frac{WYF}{NMFP}$

KLMC: thousands of lines of maintained code

NMFP: Number of function points designated for maintained software

Failure Severity & Maintenance Metrics

- Average Severity of Software System Failures (ASSSF)
 - Software failures detected during one year period

$$ASSSF = \frac{WYF}{NYF}$$

- Maintenance Repair Failure (MRepF)
 - Directly failed and repeated maintenance required

$$MRepF = \frac{RepYF}{NYF}$$

RepYF: Number of repeated software failure calls

Availability Metrics

Full Availability

All functions perform properly

$$FA = \frac{NYSerH - NYFH}{NYSerH}$$

Vital Availability

Not vital functions fail

$$Vita = \frac{NYSerH - NYVitH}{NYSerH}$$

Total Unavailability

All functions fail

$$TUA = \frac{NYTFH}{NYSerH}$$

NYSerH: Number of hours software is in service in one year

NYFH: Number of hours where at least one function is unavailable

NYVitH: Number of hours when at least one vital function is unavailable

NYTFH: Number of hours of total failure in one year

$$NYTFH > NYVitH > NYFH$$